Gated TimeMixer-XGBoost for PM2.5 Forecasting

Ngan Pham (0009-0001-1089-2626)[1] Nhi Tran (0009-0001-1047-5080)[1] Han Dong (0009-0002-2532-8099)[1] Thu Le (0009-0008-3480-8311)[1]

Faculty of Information Technology, FPT University, Ho Chi Minh City, Vietnam
Emails: tunganphamhoang@gmail.com; nhikl12006@gmail.com; donghan010106@gmail.com; thulvm@fpt.edu.vn

Abstract Accurate multi-horizon PM2.5 forecasting is difficult because hourly monitoring records combine persistence, periodicity, abrupt episodes, missing observations, site heterogeneity, and measurement-regime changes. This study evaluates a calibration-gated TimeMixer-XGBoost framework at five UK-Air stations and horizons of 1, 6, 12, 24, and 48 hours. The residual stage removes calibration-period median bias, fits horizon-specific XGBoost models with pseudo-Huber loss and bounded corrections, selects a correction scale on a later held-out calibration subset, and applies rolling calibration using only errors observable at each forecast origin. The benchmark includes persistence, seasonal-naive, direct XGBoost, DLinear, LSTM, TimeMixer, ungated robust correction, gated static correction, and gated rolling correction. Neural experiments use seeds 0-4 and a 30-epoch maximum under separate training, early-stopping, residual-fit, gate-selection, and locked-test eras. The confirmatory experiment completed five stations, five seeds, and a 30-epoch maximum. Relative to TimeMixer, the gated rolling hybrid reduced pooled MAE at all five horizons, with improvements ranging from 0.84% at 48 hours to 42.58% at 1 hour. Calibration gating suppressed 56.8% of station–seed–horizon corrections, substantially limiting the over-correction observed in the ungated ablation. Nevertheless, the hybrid won 0 of 25 comparisons against the strongest matched non-hybrid comparator; its mean relative MAE improvement under that stricter comparison was -23.33%. LSTM obtained the lowest pooled MAE (3.271). The results support gating as an effective TimeMixer correction safeguard but not a superiority or state-of-the-art claim.

Keywords: PM2.5; TimeMixer; XGBoost; multi-horizon forecasting; air quality; residual learning; time-series forecasting; reproducibility

1 Introduction

Fine particulate matter ($PM_{2.5}$) is a central air-quality indicator because long-term and short-term exposure is associated with respiratory and cardiovascular risk [1]. For city operators and public-facing dashboards, the useful question is not only whether pollution is high now, but whether it is likely to remain high over the next few hours or days. This study addresses that operational setting through multi-horizon hourly forecasting at 1, 6, 12, 24, and 48 hours.

Historical $PM_{2.5}$ forecasting is difficult for three connected reasons. First, the signal is strongly persistent at short horizons, but persistence weakens when the lead time increases. Second, daily and weekly cycles coexist with abrupt pollution episodes, producing a mixture of smooth temporal structure and sharp residual errors. Third, monitoring archives are imperfect: they contain missing observations, long outages, negative source values, and instrument-regime changes. A model can appear accurate if these problems are silently interpolated or split randomly across training and test sets.

The paper therefore evaluates a leakage-controlled, auditable forecasting pipeline rather than only a model architecture. The proposed model uses TimeMixer as a multiscale base forecaster and XGBoost as a horizon-specific residual corrector. The residual layer is deliberately conservative: it removes calibration-period median bias, trains with pseudo-Huber loss, clips extreme corrections, selects a non-negative gate on a held-out calibration subset, and updates recent bias through rolling calibration using only errors observable at the forecast origin. This design asks a practical question: can a boosted residual layer improve a neural multiscale forecaster without creating unstable over-correction?

The empirical study uses five UK-Air stations representing urban traffic, urban background, and rural background environments. All experiments use the same chronological partitions, origin-safe features, five random seeds, and a 30-epoch maximum for neural models. The final results are intentionally reported against two standards. The first asks whether the gated residual layer improves the TimeMixer base. The second, stricter standard asks whether the hybrid beats the strongest non-hybrid comparator at each station and horizon. This distinction is important: the proposed correction is useful as a safeguard for TimeMixer, but it does not support a state-of-the-art claim against the best baseline in this dataset.

The contributions are threefold. First, the study provides a reproducible PM2.5-only pipeline that records cleaning decisions instead of hiding them inside preprocessing. Second, it tests a conservative residual-correction design that can abstain when calibration evidence does not support adjustment. Third, it reports both positive and negative evidence, making clear where the hybrid helps, where simpler or recurrent baselines remain stronger, and why future improvement should focus on base-model strength and external drivers.

2 Related Work

Classical forecasting methods remain relevant because air-quality time series contain persistence and calendar structure that simple baselines can exploit [2]. Persistence and seasonal-naive forecasts are therefore not placeholders; at short horizons, they are often hard to beat. For PM2.5 specifically, previous studies show that model ranking varies with input variables, architecture, and forecast lead time [3]. This motivates horizon-by-horizon reporting rather than a single averaged score.

Neural sequence models provide stronger nonlinear temporal representations. LSTM is a common recurrent baseline for hourly PM2.5 forecasting [4, 5]. More recent long-horizon models represent temporal structure through decomposition, patches, or multiscale mixing, including N-BEATS [6], PatchTST [7], DLinear [8], Temporal Fusion Transformers [9], and TimeMixer [10]. TimeMixer is especially relevant here because it mixes information across downsampled temporal scales, which matches the coexistence of short-lag persistence and slower pollution cycles.

Tree ensembles offer a different advantage. XGBoost models nonlinear interactions among lagged values, rolling statistics, calendar variables, and measurement-regime indicators while remaining efficient on tabular features [11]. Related gradient-boosting systems such as LightGBM show the value of histogram-based tree learning for structured prediction [12]. In this paper, XGBoost is not used to replace the sequence model; it is used to correct residual errors that remain after TimeMixer. This follows the broader direct multi-step forecasting idea that different lead times can require different error models [13].

Evaluation must also respect the time-series structure. Random cross-validation can leak near-duplicate temporal contexts across train and test splits [14]. The study therefore uses chronological partitions, origin-safe feature construction, MAE as the primary metric [15], and paired evidence for dependent forecast errors [16, 17]. SHAP summaries are used only as predictive diagnostics, not causal explanations [18]. This evaluation framing is especially important for hybrid models. If the residual learner

Table 1: Station-specific data limitations identified before model fitting.

Station	Type	Available-history issue	Pre-specified treatment
MY1	Urban traffic	2020 coverage 77.81%; 69-day gap	Preserve long gap; reject affected windows
BIR_A4540	Urban traffic	Record begins in 2021	Report shorter training history
HP1	Urban background	No observations in 2016-2018	Treat as unavailable, not imputed
CHB	Rural background	2024 coverage 73.44%; 561 negatives	Mask negatives; preserve unresolved gaps
KC1	Urban background	Instrument-regime transition	Retain instrument type as context

is tuned on the same period used for final testing, it can appear to improve the base forecaster while actually learning test-specific bias. For that reason, this paper separates neural validation, residual fitting, gate selection, and locked testing into distinct chronological eras.

3 Method

The method combines data cleaning, origin-safe feature construction, multi-horizon neural forecasting, residual calibration, and locked evaluation. The goal is to preserve the evidence trail from raw monitoring records while preventing future information from entering model inputs.

3.1 Data and monitoring stations

The confirmatory dataset contains five UK-Air stations selected to represent different monitoring environments: London Marylebone Road (MY1, urban traffic), Birmingham A4540 (BIR_A4540, urban traffic), London Honor Oak Park (HP1, urban background), Chilbolton (CHB, rural background), and London North Kensington (KC1, urban background). Marylebone Road is a well-studied urban roadside site [19], and UK-Air metadata are used to verify station identity and monitoring context [20, 21, 22]. Each station is processed independently with the same schema, cleaning rules, forecast horizons, chronological boundaries, seeds, and evaluation metrics. Dataset paths and site types are fixed in `data/stations.csv`.

UK-Air exports separate date and hour-ending time fields. During ingestion, the pipeline preserves source filename, source row identity, raw timestamp text, raw concentration text, and status metadata. A source time of 24:00 is normalized to 00:00 of the following day. The target is hourly PM2.5 mass concentration in $\mu g\, m^{-3}$.

These differences imply unequal training histories. The primary analysis uses each station's legitimately available archive because that reflects the operational data available to a forecaster. A separate common-history sensitivity runner is included for reviewer checks, but the present confirmatory claims use the completed five-station primary run.

3.2 Auditable cleaning

The raw file is immutable. Cleaning creates three concentration concepts: *raw* values preserve source evidence, *observed* values contain numeric non-negative measurements, and *clean* values add only approved short-gap interpolation. Each row carries flags for invalid timestamps, missing or non-numeric source values, negative values, duplicate timestamps, inserted hourly-grid rows, interpolation, and preserved long gaps.

Table 2: Verified quality profile of the supplied London Marylebone Road record.

Audit item	Value
Raw hourly rows	87 672
Missing/non-numeric readings	8 087
Negative numeric measurements	238
Longest invalid run	1 656 h (69 d)
2020 valid coverage after negative exclusion	77.81%
Instrument transition	22 Dec 2021, 12:00 UTC
Primary short-gap interpolation limit	3 h

Linear interpolation is allowed only for internal gaps no longer than three hours. Longer gaps remain missing, and any model window crossing unresolved missingness is rejected. Instrument information is parsed from source status metadata and encoded as TEOM FDMS = 0 and BAM = 1, because measurement regime can alter the statistical distribution even when it is not causal.

3.3 Origin-safe predictors

Calendar predictors contain hour, day of week, month, weekend status, and sine/cosine encodings. Direct XGBoost uses PM2.5 lags at 1, 2, 3, 6, 12, 24, 48, 72, and 168 hours, shifted rolling summaries over 3 to 168 hours, differences, cyclic calendar variables, and instrument type. Every rolling summary is computed after a one-hour shift, so a feature attached to origin t contains values no later than $t-1$. The current value y_t remains available as the last element of the 168-hour sequence.

This origin-safety rule is treated as a pipeline invariant rather than a theorem in the main text: every sequence, lag, rolling, calendar, instrument, and missingness predictor must be measurable at or before forecast origin t, while every target must be strictly after t and remain inside its assigned chronological partition. This matters because multi-horizon windows overlap heavily. A random split would place nearly identical contexts on both sides of a boundary, whereas chronological partitions preserve the information set available at deployment.

3.4 Forecasting task

Let y_t denote hourly PM2.5. Each model uses the previous 168 hours as an origin-available multichannel sequence containing cleaned PM2.5, a missingness mask, instrument type, and cyclic calendar encodings. For horizon set $\mathcal{H} = \{1, 6, 12, 24, 48\}$, the direct task is

$$\hat{\mathbf{y}}_t = (\hat{y}_{t+1}, \hat{y}_{t+6}, \hat{y}_{t+12}, \hat{y}_{t+24}, \hat{y}_{t+48}) . \tag{1}$$

Predictions are generated simultaneously and are not recursively fed back. Direct prediction avoids recursive error accumulation and permits one residual correction function per horizon.

3.5 TimeMixer base forecaster

The implementation is a compact research adaptation of the multiscale ideas in TimeMixer rather than a line-for-line reproduction of the official benchmark code [10]. The input sequence is downsampled at several scales, each scale is decomposed into smooth and residual components, and MLP mixing blocks combine information across scale and feature dimensions. A linear head produces the five horizon forecasts, and the learned multiscale embedding is reused by the residual XGBoost stage. Huber loss, AdamW optimization, gradient clipping, and early stopping are used. Input and target standardizers are fitted on training windows only.

3.6 Gated residual XGBoost

The residual-calibration interval is divided chronologically into an earlier residual-fit subset (70%) and a later gate-selection subset (30%). For each horizon, the residual is centred by its calibration median, learned by a separate robust XGBoost model, clipped to calibration quantiles, and then scaled by a held-out gate:

$$\hat{y}^{HYB}_{t,h} = \hat{y}^{TM}_{t,h} + \gamma_h \hat{c}_{t,h} + \lambda_t \ \mathrm{median}(\mathcal{E}_{t,h}). \tag{2}$$

Here $\gamma_h \in \{0, 0.10, 0.25, 0.50, 0.75, 1\}$ is selected on the later calibration subset. The explicit choice $\gamma_h = 0$ falls back to the TimeMixer base when the correction does not improve held-out calibration MAE. The context vector contains origin-safe lag and shifted rolling features, instrument type, cyclic calendar variables, missingness state, gap length, interpolation status, and measurement-quality flags. The complete base forecast vector and latent embedding are also supplied to every horizon model.

Each residual booster uses pseudo-Huber loss, bounded correction quantiles, row and feature subsampling, and L_1/L_2 regularization. The gate γ_h is selected on $\mathcal{C}_g$, not on the test set. If no correction improves calibration MAE, $\gamma_h = 0$ falls back to the TimeMixer base. The rolling set $\mathcal{E}_{t,h}$ contains only errors from the previous 30 days whose target timestamps are no later than origin t; shrinkage prevents unstable updates when few errors are available.

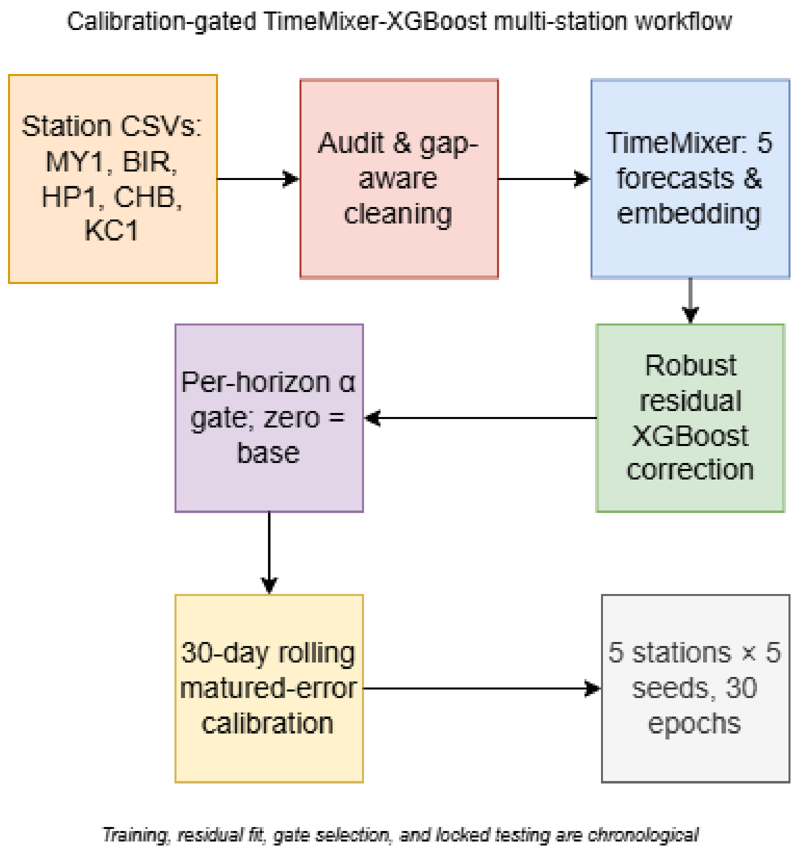

Fig. 1: End-to-end framework. TimeMixer supplies the multi-horizon base forecast; a later calibration interval fits and gates the robust residual correction before locked testing.

Table 3: Temporal partitions and permitted use.

Partition	Period	Permitted use
Training	through 31 Dec 2022	scalers; neural weights; direct XG-Boost
Neural validation	1 Jan-31 Dec 2023	early stopping only
Hybrid calibration	1 Jan-30 Jun 2024	residual fit (first 70%); gate selection (last 30%)
Locked test	after 30 Jun 2024	final paired evaluation only

3.7 Experimental setup

Four consecutive eras separate fitting, early stopping, residual calibration, and locked testing. Historical context may cross a boundary, but forecast targets must remain inside their assigned partition.

Comparators are persistence, seasonal-naive, direct XGBoost, DLinear [8], LSTM, and TimeMixer. Static and rolling hybrid variants isolate residual safeguards. Neural models use seeds 0–4, batch size 128, AdamW at 10^{-3}, early-stopping patience seven, and at most 30 epochs. MAE is primary; RMSE, sMAPE, R^2, bias, and fixed training-90th-percentile event metrics are secondary. A paired bootstrap resamples 24-hour blocks of the absolute-error difference between the hybrid and TimeMixer; negative differences favor the hybrid. Diebold-Mariano tests use absolute loss and Holm correction [16]. Same-origin ablations isolate embeddings, context, gating, and rolling calibration.

4 Results

The first empirical finding is that preprocessing choices materially change the admissible forecasting sample. At MY1, the record contains 8,087 missing or non-numeric readings, 238 negative values, and a 1,656-hour maximum invalid run. Cross-station auditing reveals additional heterogeneity: HP1 contains no observations in 2016-2018; BIR_A4540 begins in 2021; CHB has only 73.44% coverage in 2024 and contains 561 negative source values; and KC1 contains an instrument-regime transition. None of these intervals or negative values is reconstructed through long-range interpolation. Instead, invalid values remain auditable and forecast windows crossing unresolved gaps are excluded.

Figure 2 summarizes the useful temporal signal: PM2.5 is right-skewed, has repeated daily structure, and remains autocorrelated at daily and weekly lags. Figure 3 shows why this structure cannot be separated from data quality. Coverage changes by year and month, and long gaps would create artificial trends if interpolated. For this reason, the main text keeps only two EDA figures; the single-purpose EDA plots are retained as supplementary outputs generated by the same visualization code.

Figure 4 is the main effectiveness figure. It shows that the hybrid does not beat the strongest comparator in any of the 25 station-horizon cells. The result exporter accepts only complete gated-v2 runs covering all five stations, seeds 0–4, and the 30-epoch configuration, so the reported comparison is not based on a cherry-picked single run.

The gated rolling hybrid won 0 of 25 matched station–horizon comparisons. Its mean relative MAE improvement against the strongest non-hybrid comparator was -23.33%.

Table 4 explains which models are strongest by horizon. **XGBoost** is strongest at 1 hour, while **LSTM** is strongest from 6 to 48 hours. The hybrid improves its intended TimeMixer base at every horizon, reducing pooled MAE by 42.58% at 1 hour and by

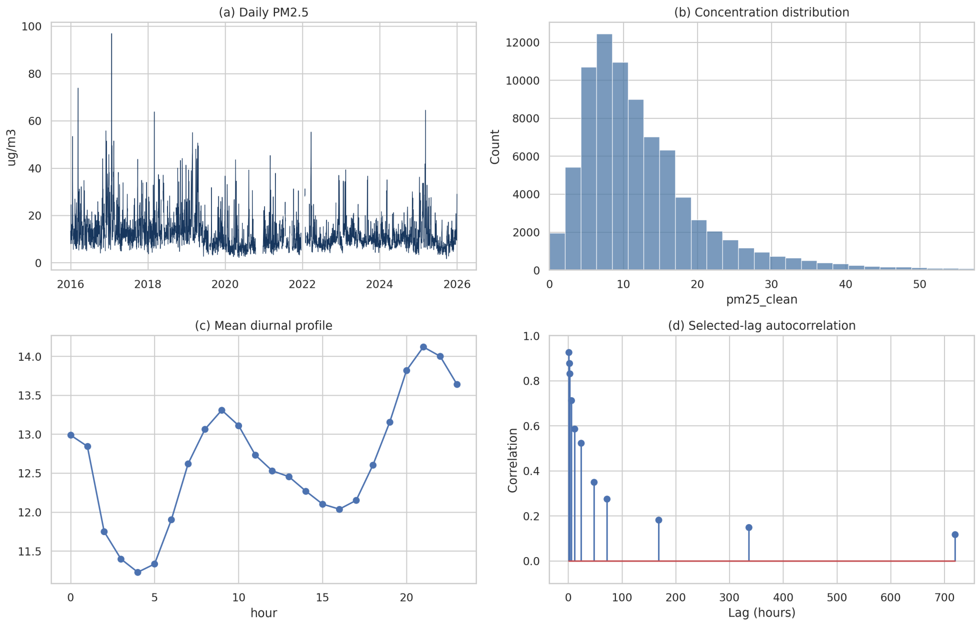

Fig. 2: EDA overview: daily PM2.5, distribution, diurnal profile, and autocorrelation.

Fig. 3: Data-quality audit: coverage, monthly missingness, gap duration, and the preserved long outage.

smaller but positive margins from 6 to 48 hours. However, this is a narrower result than overall superiority: the hybrid pooled MAE is 3.708, compared with 4.078 for TimeMixer and **3.271 for LSTM**.

The ablation study is intentionally same-origin and chronological. It distinguishes the TimeMixer base, ungated robust correction, calibration-gated static correction, rolling correction, and reduced-feature variants. This allows the paper to separate nonlinear residual fit, protection against over-correction, and adaptation to recent drift.

The ablation identifies over-correction as the main failure mode of the ungated residual learner. At horizons 12, 24, and 48 hours, its pooled MAE was 6.020, 6.420, and 6.632, respectively. Calibration gating reduced these values to 4.144, 4.320, and 4.391, and rolling calibration further reduced them to 4.075, 4.264, and 4.359. The gate selected zero correction in 56.8% of all station-seed-horizon fits. Mean gate weight

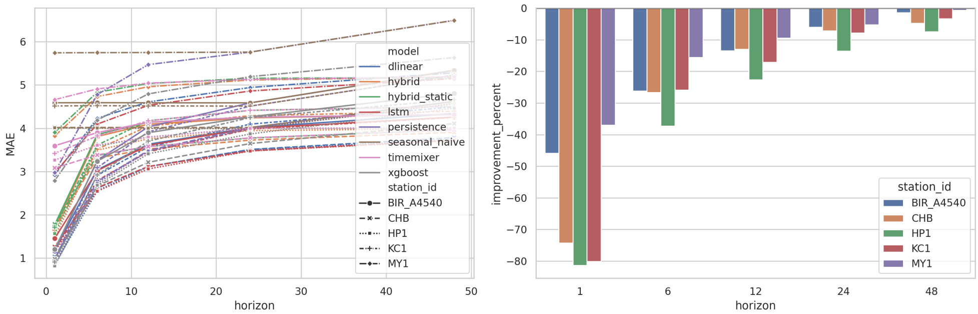

Fig. 4: Five-seed multi-station effectiveness. Each row compares the hybrid with the strongest non-hybrid comparator at the same station and horizon; values below zero mean that the comparator remains better.

Table 4: Locked-test MAE pooled over station–seed runs. Parentheses show dispersion across both stations and seeds; station-specific comparisons are retained in the machine-readable results.

Model	1 h	6 h	12 h	24 h	48 h
dlinear	1.390 (0.791)	3.070 (0.621)	3.607 (0.558)	4.014 (0.537)	4.329 (0.574)
hybrid	2.070 (0.902)	3.773 (0.519)	4.075 (0.502)	4.264 (0.489)	4.359 (0.472)
hybrid_static	2.149 (0.906)	3.853 (0.531)	4.144 (0.515)	4.320 (0.494)	4.391 (0.460)
lstm	1.604 (0.708)	**3.016 (0.578)**	**3.555 (0.541)**	**3.959 (0.519)**	**4.220 (0.536)**
persistence	1.394 (0.816)	3.324 (0.769)	4.076 (0.748)	4.574 (0.655)	5.251 (0.709)
seasonal_naive	4.576 (0.645)	4.577 (0.648)	4.574 (0.650)	4.574 (0.655)	5.251 (0.709)
timemixer	3.605 (0.577)	3.925 (0.540)	4.146 (0.518)	4.316 (0.483)	4.396 (0.469)
xgboost	**1.341 (0.751)**	3.140 (0.576)	3.804 (0.559)	4.244 (0.540)	4.735 (0.519)

was 0.930 at 1 hour but no more than 0.158 at longer horizons, showing that strong residual correction was supported primarily at short lead time.

SHAP and bootstrap diagnostics are retained in the supplementary generated figures rather than the main text. This avoids repeating the same message with several plot types: the residual layer improves TimeMixer after gating, but it does not become the best overall forecaster.

5 Discussion

Across all station–seed–horizon rows, the hybrid pooled MAE was 3.708, compared with 4.078 for TimeMixer and 3.271 for LSTM. The held-out gate selected zero correction in 56.8% of fits and had mean alpha 0.24, quantifying how often fallback protection was needed. Because gate selection did not access the locked test, the remaining gap reflects generalization rather than post-hoc test selection. The method therefore achieved its narrow goal of improving TimeMixer while failing the stricter goal of beating the best model available at each station and horizon.

5.1 Interpretation of effectiveness

The results explain both why the proposed method is reasonable and why it is not yet globally effective. The TimeMixer base captures multiscale temporal structure, but its residuals are not equally learnable at all horizons. At 1 hour, recent lag information remains highly informative, so the residual gate often permits a large correction. At longer horizons, the gate usually shrinks toward zero because residual predictions become less reliable. This behavior is desirable: a residual model that is allowed to correct every forecast can produce large negative transfer, as shown by the ungated ablation.

The strongest competing models also clarify the limitation. **XGBoost** is best at 1 hour because origin-available lags and rolling statistics almost directly describe the next-hour state. **LSTM** is best at longer horizons because it learns a smoother sequence representation than the compact TimeMixer implementation used here. In other words, the residual layer repairs some TimeMixer errors, but it cannot fully compensate for a weaker base trajectory. The method is therefore best interpreted as a calibrated residual safeguard rather than a new state-of-the-art forecaster.

5.2 Limitations

Several threats remain. Overlapping forecast windows create serial dependence, so paired block evidence is more appropriate than independent-sample tests, but it does not remove all dependence. Five UK stations are broader than a single-site study but do not establish transfer to unseen industrial, suburban, or non-UK environments. Unequal station histories also matter: HP1 lacks 2016–2018, BIR_A4540 begins in 2021, CHB has reduced 2024 coverage, and KC1 contains an instrument transition. The common-history sensitivity code is included to examine this issue, but only completed primary results are used for the present claims.

The study is also limited by its PM2.5-only design. Historical concentrations do not include forecast-available meteorology, traffic intensity, co-pollutants, transport patterns, or local event information. These omitted drivers are likely important during abrupt pollution episodes. Finally, instrument indicators are useful regime markers but should not be interpreted as causal explanations of pollution concentration.

5.3 Future work

Future work should therefore test stronger base forecasters, add only variables known at the forecast origin, evaluate leave-one-station-out transfer, and use regime-specific calibration for instrument changes and seasonal drift. A useful next experiment is to compare the PM2.5-only setting with a meteorology-assisted setting under the same locked-test protocol. The workflow should still preserve raw audit fields, abstain across unresolved long gaps, and keep the locked test independent of model tuning.

6 Conclusion

This paper evaluated a gated TimeMixer-XGBoost approach for multi-horizon hourly PM2.5 forecasting. The study contributes a complete, auditable forecasting workflow: raw source values are preserved, invalid and negative measurements are flagged, long gaps are not artificially interpolated, and features are constructed so that every forecast uses only information available at its origin. The experimental design covers five monitoring stations, five forecast horizons, five neural seeds, and a 30-epoch maximum under chronological training, validation, calibration, and locked-test partitions.

The main result is mixed but informative. The gated residual layer improves the TimeMixer base at every horizon and greatly reduces the over-correction seen in ungated residual learning. This supports the proposed de-biasing, robust loss, held-out gating, and rolling calibration strategy as a useful safeguard. However, the hybrid wins 0 of 25 station-horizon comparisons against the strongest non-hybrid comparator, and **LSTM** obtains the best pooled MAE overall. The correct conclusion is therefore not that the hybrid is state-of-the-art, but that calibrated residual correction can make a multiscale neural forecaster safer and more competitive.

For a student AI project, this distinction is valuable. The work does not hide negative results; instead, it explains why they occur, identifies the horizons where residual correction is most useful, and leaves a reproducible path for future improvement. The next step is to strengthen the base forecaster and add forecast-available external drivers while keeping the same leakage-safe evaluation protocol.

Overall, the study shows that a hybrid architecture should be judged not by whether it sounds more advanced, but by whether each added component improves out-of-sample behavior under fair comparison. In this dataset, the gated residual layer is justified as a controlled calibration mechanism, while LSTM and XGBoost remain important reference points for future model development.

Data and Code Availability

The repository contains data-processing and modeling code, tests, figures, experiment outputs, and manuscript source. UK-Air attribution and licensing conditions must be preserved.

Conflict of Interest

The author declares no conflict of interest.

References

[1] World Health Organization. WHO global air quality guidelines: Particulate matter (PM2.5 and PM10), ozone, nitrogen dioxide, sulfur dioxide and carbon monoxide. World Health Organization, 2021.

[2] Rob J. Hyndman and George Athanasopoulos. *Forecasting: Principles and Practice*. OTexts, Melbourne, Australia, 3rd edition, 2021.

[3] Jie Yang, Rui Yan, Mingyue Nong, Jiaqiang Liao, Feipeng Li, and Wei Sun. Pm2.5 concentrations forecasting in beijing through deep learning with different inputs, model structures and forecast time. *Atmospheric Pollution Research*, 12(9):101168, 2021.

[4] Sepp Hochreiter and Jürgen Schmidhuber. Long short-term memory. *Neural Computation*, 9(8):1735–1780, 1997.

[5] Yun Bai, Bo Zeng, Chuan Li, and Jin Zhang. An ensemble long short-term memory neural network for hourly pm2.5 concentration forecasting. *Chemosphere*, 222:286–294, 2019.

[6] Boris N. Oreshkin, Dmitri Carpov, Nicolas Chapados, and Yoshua Bengio. N-beats: Neural basis expansion analysis for interpretable time series forecasting. In *International Conference on Learning Representations*, 2020.

[7] Yuqi Nie, Nam H. Nguyen, Phanwadee Sinthong, and Jayant Kalagnanam. A time series is worth 64 words: Long-term forecasting with transformers. In *International Conference on Learning Representations*, 2023.

[8] Ailing Zeng, Muxi Chen, Lei Zhang, and Qiang Xu. Are transformers effective for time series forecasting? In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 37, pages 11121–11128, 2023.

[9] Bryan Lim, Sercan Ö. Arik, Nicolas Loeff, and Tomas Pfister. Temporal fusion transformers for interpretable multi-horizon time series forecasting. *International Journal of Forecasting*, 37(4):1748–1764, 2021.

[10] Shiyu Wang, Haixu Wu, Xiaoming Shi, Tengge Hu, Huakun Luo, Lintao Ma, James Y. Zhang, and Jun Zhou. Timemixer: Decomposable multiscale mixing for time series forecasting. In *The Twelfth International Conference on Learning Representations*, 2024.

[11] Tianqi Chen and Carlos Guestrin. Xgboost: A scalable tree boosting system. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 785–794, 2016.

[12] Guolin Ke, Qi Meng, Thomas Finley, Taifeng Wang, Wei Chen, Weidong Ma, Qiwei Ye, and Tie-Yan Liu. Lightgbm: A highly efficient gradient boosting decision tree. In *Advances in Neural Information Processing Systems*, volume 30, 2017.

[13] Souhaib Ben Taieb, Gianluca Bontempi, Amir F. Atiya, and Antti Sorjamaa. A review and comparison of strategies for multi-step ahead time series forecasting based on the nn5 forecasting competition. *Expert Systems with Applications*, 39(8):7067–7083, 2012.

[14] Christoph Bergmeir, Rob J. Hyndman, and Bonsoo Koo. A note on the validity of cross-validation for evaluating autoregressive time series prediction. *Computational Statistics & Data Analysis*, 120:70–83, 2018.

[15] Cort J. Willmott and Kenji Matsuura. Advantages of the mean absolute error (MAE) over the root mean square error (RMSE) in assessing average model performance. *Climate Research*, 30(1):79–82, 2005.

[16] Francis X. Diebold and Roberto S. Mariano. Comparing predictive accuracy. *Journal of Business & Economic Statistics*, 13(3):253–263, 1995.

[17] Whitney K. Newey and Kenneth D. West. A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. *Econometrica*, 55(3):703–708, 1987.

[18] Scott M. Lundberg and Su-In Lee. A unified approach to interpreting model predictions. In *Advances in Neural Information Processing Systems*, volume 30, pages 4765–4774, 2017.

[19] Adams A. Kamara and Roy M. Harrison. Analysis of the air pollution climate of a central urban roadside supersite: London, Marylebone Road. *Atmospheric Environment*, 258:118479, 2021.

[20] Department for Environment, Food & Rural Affairs. Site information for London Marylebone Road (uka00315). UK AIR Air Information Resource. Accessed 5 July 2026.

[21] Department for Environment, Food & Rural Affairs. CSV Data Files for London N. Kensington Monitoring Site. UK-AIR Air Information Resource, 2026.

[22] Department for Environment, Food & Rural Affairs. Site Information for London Honor Oak Park (UKA00656). UK-AIR Air Information Resource, 2026.

Appendix: Core Notation

Table 5: Core notation used in the forecasting formulation.

Symbol	Meaning
y_t	observed hourly PM2.5 at time t
L	input history length, fixed at 168 h
$\mathcal{H}$	direct horizon set $\{1, 6, 12, 24, 48\}$ h
$\mathbf{x}_{t-L+1:t}$	origin-available multichannel sequence
$\hat{\mathbf{y}}_t^{TM}$	TimeMixer base forecast vector
$\mathbf{e}_t$	flattened multiscale TimeMixer embedding
$\mathbf{z}_t$	calendar, instrument, and missingness context
$r_{t,h}$	TimeMixer residual at horizon h
$g_h(\cdot)$	XGBoost residual model for horizon h
γ_h	held-out calibration correction gate for horizon h